

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8492**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	Block-401	Seat No:	2022600065
Course Section:	1				

L_29_522_986_L_1505_38296



Name of the candidate. **Tash Vora**

Candidate's UID number:

2022600065

Examination class room number:

401

Invigilator's signature with date:

8/10/25

(To be filled in by the candidate)

EXAMINATION: **BE BTech MSE**

(For example FY MCA/FY BTECH)

PROGRAM: **CSE - AIML**

DAY AND DATE: **8/10/2025, Wednesday**

COURSE NAME: **Deep Learning**

SEMESTER : **7**
I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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(Begin answer for each question on a new page)

1a

$$f(w) = \frac{1}{2} (xw - y)^2$$

let the loss function of $f(w)$ be $L(w)$

~~$$\frac{\partial f(w)}{\partial w} = \frac{1}{2} \frac{\partial}{\partial w} (x^2 w^2 - 2xwy + y^2)$$

$$= \frac{1}{2} (2wx - 2xy)$$~~

~~$$= wx - xy$$~~

~~$$= x(w - y)$$~~

Here, w is weight matrix, and y is the target matrix, with x being the input matrix

~~$$\frac{\partial f}{\partial w} = \frac{1}{2} \frac{\partial}{\partial w} (x^2 w^2 - 2xwy + y^2)$$

$$= wx^2 - 2xy$$~~

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1b)

The ~~square~~ expected square error of a predictor is given by following:
 $(y - \hat{y})^2$ ($\hat{y}$ is predicted target)

For $y = wx + b$, where $b = \text{bias}$,

the loss will be calculated using
 $(y - (wx + b))^2$

$$= (y - wx - b)^2 = (y - wx - b)^2$$

Taking partial derivative ^(w.r.t. x and b) we get

$$= -2wx - 2b \quad (\text{This is the gradient by the})$$

Thus, we can infer that in 1), the ~~loss~~ b^2 term b^2 is independent of input, thus the bias-variance decomposition of expected square error:

When the model complexity is too high, the model tends to tune the bias b to a very high sensitivity relation to training data, such that, on unseen test data, the bias-variance of ~~the~~ is much more higher than that of the training data.

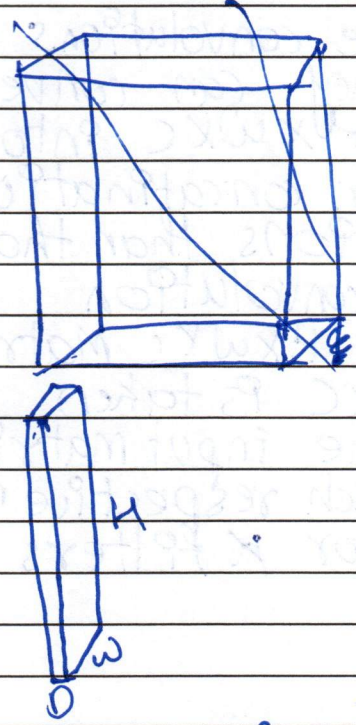
Also, the model tends to memorize the training data due to the high sensitivity.



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2a



Here, k filters are used of size $F \times F$, so the convolutions of $F \times F$ will be applied for k times for the $H \times W \times K$ matrix.
Let's consider a 4×4 matrix with 2×2 filter size convolutions (with $S=1$ and $P=0$)

a	b	c	d
e	f	g	h
i	j	k	l
m	n	o	p

w	x
y	z

the resultant matrix will be,

$aw+bx+cy+dz$ $bw+cx+fy+gz$ $cw+dx+ty+hz$
 $ew+fx+gy+hz$ $fw+gx+py+tz$ $gw+hx+ty+tz$
 $pw+qx+ym+zn$ $qw+rx+yn+zo$ $kw+xt+oy+pz$
The resultant matrix is of size 3×3

∴ The formula formed would be

$$\frac{H-F+2P+1}{S}$$

So, for the matrix of $H \times W \times K$, the following formula will be applied for the $H \times W$ matrix (k times)

Thus, for a standard 2D convolution of a $H \times W \times K$ matrix, the computational complexity is $(\frac{H-F+2P+1}{S}) \times K$



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2a

Depthwise separable convolutions are critical since they can convert a 3D matrix of $H \times W \times C$ into a 2D matrix after concatenations with fewer computations than that of a standard 2D convolution. So, suppose for a $H \times W \times C$ matrix, if the filter size $F \times F \times C$ is taken, each $H \times W$ of the input matrix will be convolved by each respective $F \times F$ matrix without the need for C filters.



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$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \end{pmatrix} y$$

$$\begin{pmatrix} 0 & 0 & 0 \end{pmatrix} h$$

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 \end{pmatrix} x$$

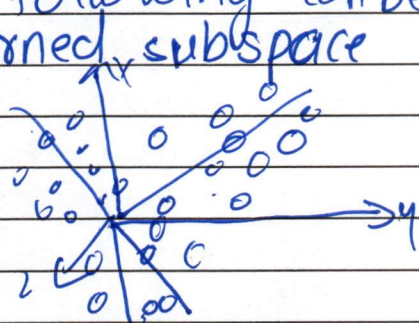
Let x be the input matrix, h be the hidden layer and y be the target matrix. The following network represents an undercomplete autoencoder.

Here, it's given that it's a linear autoencoder which means $h = (Wx + b)$

It's also given that it's trained with an MSE $\therefore L(h) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{x}_i)^2$ ($\hat{x}_i$ is predicted by h)

In PCA dimensional reductionality, a similar training objective is performed, where an input dimension needs to be reduced to n dimensions.

For example, if a 5-dim input x has to be reduced to 3-dimension with PCA, the following can be inferred about the learned subspace.



(# a sample visualization of dimensionality reduction in PCA)

Thus, in the following learned subspace, it can be inferred that PCA is using a linear function and the loss of.

Thus, linear undercomplete Autoencoder trained with MSE learns subspace equivalent to PCA.

In an overcomplete autoencoder, the $\dim(h) \geq \dim(x)$. So with a linear mapping from $x \rightarrow h$ and $h \rightarrow y$, it's expected due to nature of MSE and minimization of the error objective that the subspace will learn to copy the inputs of x for the error to be 0. Thus, to prevent the learning of identity function, regularization is used, where the subspace intentionally avoids to learn identity function.

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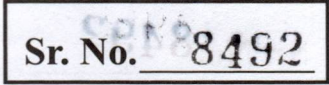


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